CADD INTERNSHIP REPORT

Internship Title: Computer-Aided Drug Design (CADD) Internship

Student Name: Aditi Gupta

College/University Name: Gautam Buddha University

Internship Organization Name: BioTechTrek

Internship Duration: 8 Week

Guide/Mentor Name: Ms. Mansi

Submission Date: 10 July 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **BioTechTrek** for providing me with the opportunity to undertake this two-month internship in **Computer-Aided Drug Design (CADD)**.

I am especially thankful to my mentor, **Ms. Mansi**, for her constant guidance, encouragement, and valuable insights throughout the internship. Her support greatly enhanced my understanding of computational drug discovery and bioinformatics.

I also extend my heartfelt thanks to all the trainers and team members at BioTechTrek for sharing their knowledge and providing practical exposure to various CADD tools and techniques.

Finally, I am grateful to everyone who directly or indirectly contributed to the successful completion of this internship and this report.

ABSTRACT

Comparative Computational Evaluation of *Moringa oleifera* Phytochemicals and FDA-Approved ACE Inhibitors Against Human Angiotensin-Converting Enzyme (ACE) Using Molecular Docking, ADMET Prediction and Molecular Dynamics Simulation

Hypertension is one of the leading causes of cardiovascular diseases worldwide. Angiotensin-Converting Enzyme (ACE) plays a crucial role in regulating blood pressure through the Renin–Angiotensin–Aldosterone System (RAAS), making it an important therapeutic target for antihypertensive drug development. Although several FDA-approved ACE inhibitors such as Captopril and Lisinopril are clinically effective, their long-term use may be associated with adverse effects. Therefore, natural phytochemicals have emerged as promising alternatives due to their diverse biological activities and comparatively lower toxicity.

The present computational study aimed to compare selected phytochemicals of *Moringa oleifera* with FDA-approved ACE inhibitors using Computer-Aided Drug Design (CADD) techniques. The three-dimensional crystal structure of Human ACE (PDB ID: **1O86**) was obtained from the Protein Data Bank, while phytochemicals were retrieved from the PubChem database. Protein and ligand structures were prepared using PyMOL, Open Babel, and PyRx prior to molecular docking using AutoDock Vina. Docking interactions were visualized using Discovery Studio Visualizer. Drug-likeness and pharmacokinetic properties were evaluated using SwissADME and pkCSM. The stability of the protein–ligand complex was further investigated through Molecular Dynamics (MD) simulation using GROMACS 2025.4.

The MD trajectory was analysed using Root Mean Square Deviation (RMSD), Root Mean Square Fluctuation (RMSF), Radius of Gyration (Rg), Hydrogen Bond analysis, and Solvent Accessible Surface Area (SASA). The RMSD remained within **0.082–0.122 nm**, while the Radius of Gyration remained approximately **2.39 nm**, indicating excellent structural stability. RMSF values demonstrated minimal residue fluctuations, and stable hydrogen bonding together with nearly constant SASA values confirmed the compactness and stability of the protein throughout the simulation.

Overall, the computational analyses demonstrated that selected *Moringa oleifera* phytochemicals exhibit promising interactions with ACE and may serve as potential lead compounds for further experimental validation. The study also provided practical exposure to molecular docking, ADMET prediction, and molecular dynamics simulation using modern computational drug discovery tools.

Keywords: Computer-Aided Drug Design, Molecular Docking, Angiotensin-Converting Enzyme, *Moringa oleifera*, Molecular Dynamics Simulation, ADMET Prediction, GROMACS, AutoDock Vina.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ACE	Angiotensin-Converting Enzyme
CADD	Computer-Aided Drug Design
MD	Molecular Dynamics
ADMET	Absorption, Distribution, Metabolism, Excretion and Toxicity
RMSD	Root Mean Square Deviation
RMSF	Root Mean Square Fluctuation
Rg	Radius of Gyration
SASA	Solvent Accessible Surface Area
PDB	Protein Data Bank
RAAS	Renin–Angiotensin–Aldosterone System
VDW	Van der Waals
H-bond	Hydrogen Bond
GI	Gastrointestinal
BBB	Blood–Brain Barrier
QSAR	Quantitative Structure–Activity Relationship

1. INTRODUCTION

Hypertension is a major public health concern affecting millions of people worldwide and is considered one of the primary risk factors for cardiovascular diseases, stroke, chronic kidney disease, and heart failure. According to the World Health Organization (WHO), hypertension contributes significantly to global morbidity and mortality. Effective management of blood pressure is therefore essential for reducing cardiovascular complications.

The Renin–Angiotensin–Aldosterone System (RAAS) plays a central role in regulating blood pressure and electrolyte balance. Angiotensin-Converting Enzyme (ACE) is a zinc-dependent metalloprotease responsible for converting Angiotensin I into the potent vasoconstrictor Angiotensin II while simultaneously degrading bradykinin, a vasodilator peptide. Consequently, inhibition of ACE results in vasodilation, decreased blood pressure, and reduced cardiovascular risk.

Several synthetic ACE inhibitors, including Captopril and Lisinopril, have been successfully employed in the treatment of hypertension. However, prolonged administration may produce adverse effects such as persistent cough, hyperkalemia, hypotension, and renal complications. These limitations have encouraged researchers to investigate natural compounds possessing ACE inhibitory activity with improved safety profiles.

Moringa oleifera, commonly referred to as the "Miracle Tree," is well known for its rich phytochemical composition. The plant contains numerous bioactive flavonoids and phenolic

compounds such as Quercetin, Kaempferol, Rutin, Isoquercetin, Astragalin, and Chlorogenic Acid, which exhibit antioxidant, anti-inflammatory, antimicrobial, and cardioprotective properties. These phytochemicals are increasingly being investigated as potential antihypertensive agents through computational drug discovery approaches.

Computer-Aided Drug Design (CADD) integrates structural biology, bioinformatics, cheminformatics, and molecular modeling to accelerate drug discovery. Molecular docking predicts the binding affinity and interaction pattern between ligands and target proteins, while ADMET analysis evaluates pharmacokinetic behavior and drug-likeness. Molecular Dynamics (MD) simulation further validates the stability of protein–ligand complexes under dynamic physiological conditions.

In this project, a comparative computational evaluation of selected *Moringa oleifera* phytochemicals and FDA-approved ACE inhibitors was carried out using molecular docking, ADMET prediction, and molecular dynamics simulation to identify promising lead compounds for future antihypertensive drug development.

TABLE OF CONTENTS

1. Introduction
   - 1.1About the Organization
   - 1.2 About the Internship
   - 1.3 Introduction to Computer-Aided Drug Design (CADD)
2. Learning Objectives
3. Tools and Software Used
4. Methodology / Workflow
5. Project Work
6. Results and Discussion
7. Skills Acquired
8. Challenges Faced and Solutions
9. Conclusion
10. References
11. Appendix (if applicable)

1. Introduction

1.1 About the Organization

BioTechTrek is a biotechnology training organization that provides industry-oriented learning programs in biotechnology, bioinformatics, computational biology, and drug discovery. The organization focuses on bridging the gap between academic knowledge and practical skills by offering hands-on training using modern computational tools and software.

Its major areas of work include molecular docking, molecular dynamics simulation, virtual screening, QSAR, ADMET prediction, bioinformatics, and computer-aided drug design. Through expert guidance and practical projects, BioTechTrek helps students gain industry-

relevant experience and prepares them for research and careers in biotechnology and pharmaceutical sciences.

1.2 About the Internship

This internship was conducted over a duration of **8 week** in an **online** mode. The primary objective of the internship was to provide practical exposure to **Computer-Aided Drug Design (CADD)** and familiarize students with the computational techniques used in modern drug discovery.

The training included protein and ligand preparation, molecular docking, virtual screening, ADMET prediction, molecular dynamics simulation, and the interpretation of computational results using various bioinformatics software and databases.

1.3 Introduction to Computer-Aided Drug Design (CADD)

Computer-Aided Drug Design (CADD) is a computational approach used to identify, design, and optimize potential drug molecules. It integrates bioinformatics, cheminformatics, structural biology, and molecular modeling to accelerate the drug discovery process while reducing the time and cost associated with traditional experimental methods.

CADD is widely used for target identification, virtual screening, molecular docking, lead optimization, pharmacokinetic prediction, and toxicity assessment. It enables researchers to predict how small molecules interact with biological targets before laboratory testing.

In modern pharmaceutical research, CADD has become an indispensable tool because it improves the efficiency of lead identification, minimizes experimental failures, supports precision medicine, and accelerates the development of safer and more effective therapeutic agents.

2. Learning Objectives

- Understand the drug discovery pipeline and the role of Computer-Aided Drug Design (CADD) from target identification to lead optimization.
- Retrieve and prepare protein structures and identify druggable binding sites or pockets.
- Understand molecular representations and file formats (SMILES, InChI, SDF, MOL2, and PDB) and perform file interconversion using RDKit and OpenBabel.
- Learn molecular descriptor calculation, drug-likeness filters (Lipinski's Rule of Five and Veber's Rule), and fingerprint-based similarity analysis.
- Perform docking protocol validation using the redocking method and RMSD analysis.
- Learn QSAR-guided virtual screening to prioritize natural compound libraries.
- Perform molecular docking using AutoDock Vina and analyze binding energies and protein-ligand interactions.
- Predict ADMET properties and drug-likeness using SwissADME and pkCSM.
- Learn the principles of Molecular Dynamics (MD) simulation, system preparation, trajectory analysis, and visualization using GROMACS.

- Interpret binding free energy and identify the most promising lead compounds.
- Develop the ability to compile computational findings into a structured internship report and professional project presentation.

3. Tools and Software Used

- Protein Data Bank (PDB)
- PubChem
- ChEMBL
- PyMOL
- Discovery Studio Visualizer
- AutoDock Vina
- PyRx
- SwissADME
- pkCSM
- GROMACS
- RDKit
- OpenBabel
- PaDEL Descriptor
- Python (Scikit-learn)
- Microsoft Excel

4. Methodology / Workflow

The internship followed a systematic workflow commonly used in Computer-Aided Drug Design (CADD).

1. Selection of the therapeutic target protein.
2. Retrieval of the three-dimensional protein structure from the Protein Data Bank (PDB).
3. Collection and preparation of ligand molecules from public chemical databases.
4. Protein and ligand preparation for molecular docking.
5. Validation of the docking protocol through redocking and RMSD analysis.
6. Molecular docking using AutoDock Vina.
7. Analysis of binding affinity and protein–ligand interactions.
8. Drug-likeness assessment using Lipinski's Rule of Five and Veber's Rule.
9. ADMET prediction using SwissADME and pkCSM.
10. Molecular Dynamics simulation and trajectory analysis using GROMACS.
11. Comparison and ranking of lead compounds based on docking, ADMET, and molecular dynamics results.
12. Interpretation of results and preparation of the final internship report and presentation.

5. PROJECT WORK

5.1 Project Title

Comparative Computational Evaluation of *Moringa oleifera* Phytochemicals and FDA-Approved ACE Inhibitors Against Human Angiotensin-Converting Enzyme (ACE) Using Molecular Docking, ADMET Prediction and Molecular Dynamics Simulation

5.2 Project Objective

Problem Statement

Hypertension is one of the most prevalent cardiovascular disorders worldwide and is a major cause of heart disease, stroke, and kidney failure. The Angiotensin-Converting Enzyme (ACE) plays a vital role in the Renin–Angiotensin–Aldosterone System (RAAS) by converting Angiotensin I into Angiotensin II, a potent vasoconstrictor responsible for increasing blood pressure. Although FDA-approved ACE inhibitors such as Captopril and Lisinopril are widely used for hypertension management, prolonged use may cause adverse effects including dry cough, hypotension, hyperkalemia, and renal dysfunction. Therefore, there is an increasing need to identify natural compounds with ACE inhibitory potential that may exhibit improved safety and efficacy.

Natural phytochemicals present in *Moringa oleifera* possess diverse pharmacological properties, including antioxidant, anti-inflammatory, cardioprotective, and antihypertensive activities. However, their interaction with ACE requires systematic computational evaluation before experimental validation.

Aim of the Project

The primary aim of this project was to evaluate the inhibitory potential of selected *Moringa oleifera* phytochemicals against Human Angiotensin-Converting Enzyme (ACE) using Computer-Aided Drug Design (CADD) techniques, and to compare their performance with FDA-approved ACE inhibitors through molecular docking, ADMET prediction, and molecular dynamics simulation.

Objectives

- Retrieve the three-dimensional structure of Human ACE from the Protein Data Bank (PDB).
- Collect phytochemicals of *Moringa oleifera* and FDA-approved ACE inhibitors from PubChem.
- Prepare protein and ligand structures for computational analysis.
- Perform molecular docking using AutoDock Vina through PyRx.
- Analyze protein–ligand interactions using Discovery Studio Visualizer.
- Validate the docking protocol through redocking and RMSD analysis.
- Evaluate drug-likeness and pharmacokinetic properties using SwissADME and pkCSM.
- Perform Molecular Dynamics simulation using GROMACS 2025.4.
- Analyze trajectory stability through RMSD, RMSF, Radius of Gyration, Hydrogen Bond, and SASA analyses.
- Compare the computational performance of phytochemicals with standard ACE inhibitors.

5.3 Description of Data

The study utilized both protein and ligand datasets collected from publicly available databases.

Target Protein

Parameter	Details
Protein Name	Human Angiotensin-Converting Enzyme (ACE)
PDB ID	**1O86**
Source	Protein Data Bank (RCSB PDB)
Organism	*Homo sapiens*
Resolution	X-ray Crystal Structure
File Format	PDB

The crystal structure of Human ACE was downloaded from the Protein Data Bank and prepared by removing water molecules, heteroatoms, and co-crystallized ligands using PyMOL before molecular docking.

Ligands Used

Two categories of ligands were selected.

A. Moringa oleifera Phytochemicals

- Kaempferol
- Quercetin
- Rutin
- Isoquercetin
- Astragalin
- Chlorogenic Acid

B. FDA-Approved ACE Inhibitors

- Captopril
- Lisinopril

The ligand structures were downloaded from the PubChem database in SDF format and converted into PDB format using Open Babel prior to docking.

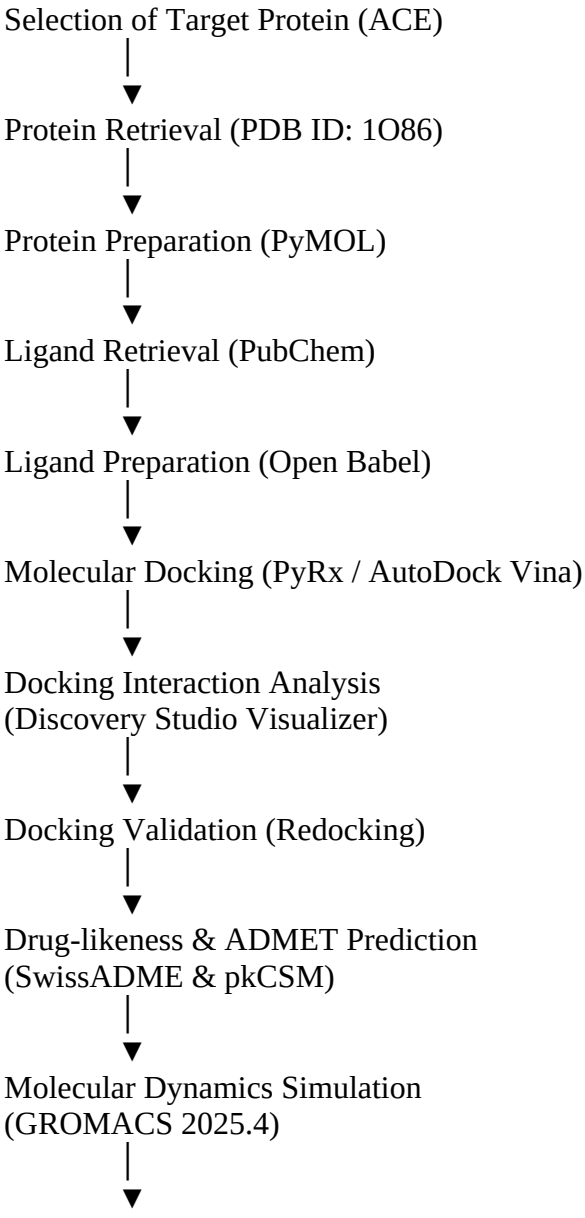
Software and Databases Used

Software/Database	Purpose
RCSB PDB	Protein Retrieval
PubChem	Ligand Retrieval

PyMOL	Protein Preparation
Open Babel	File Conversion
PyRx	Molecular Docking
AutoDock Vina	Docking Engine
Discovery Studio Visualizer	Interaction Analysis
SwissADME	Drug-likeness Prediction
pkCSM	ADMET Prediction
GROMACS 2025.4	Molecular Dynamics Simulation
Ubuntu (WSL)	MD Simulation Environment
Microsoft Excel	Data Analysis

5.4 Workflow Followed

The computational workflow adopted during this internship is illustrated below.



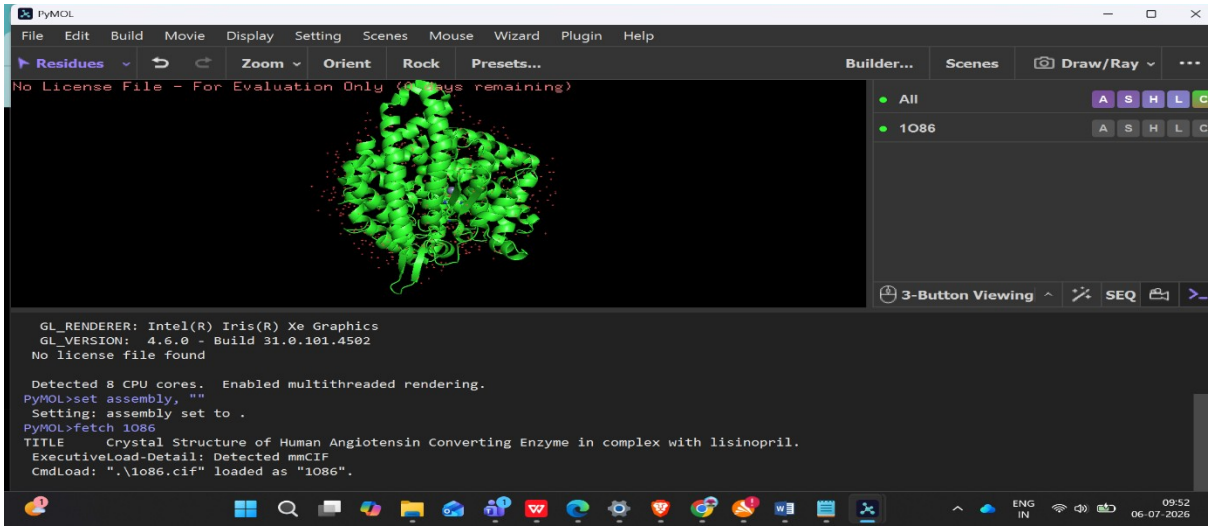
Trajectory Analysis
(RMSD, RMSF, Rg, H-Bonds, SASA)
↓
Final Interpretation and Conclusion

Detailed Workflow

1. The three-dimensional crystal structure of Human ACE (PDB ID: 1O86) was downloaded from the Protein Data Bank.
2. The protein structure was cleaned by removing water molecules and unnecessary heteroatoms using PyMOL.
3. Selected phytochemicals of *Moringa oleifera* and FDA-approved ACE inhibitors were downloaded from PubChem.
4. Ligands were converted from SDF to PDB format using Open Babel and prepared for docking.
5. Molecular docking was performed using AutoDock Vina implemented in PyRx.
6. Docking interactions were analyzed using Discovery Studio Visualizer to identify hydrogen bonds, hydrophobic interactions, and active-site residues.
7. The docking protocol was validated by redocking the native ligand and calculating RMSD.
8. Drug-likeness and pharmacokinetic properties were predicted using SwissADME and pkCSM.
9. Molecular Dynamics simulation was carried out using GROMACS 2025.4 under Ubuntu (WSL), including energy minimization, NVT equilibration, NPT equilibration, and production MD.
10. The MD trajectory was analyzed using RMSD, RMSF, Radius of Gyration, Hydrogen Bond analysis, and SASA to evaluate structural stability.
11. The computational results were interpreted to compare the phytochemicals with FDA-approved ACE inhibitors.

5.5 Results Obtained

Figure 5.1: Three-dimensional structure of ACE protein (PDB ID: 1O86) with the docked ligand visualized in PyMOL



The prepared ACE protein structure was visualized in PyMOL after protein preparation. The ligand was observed to occupy the active binding pocket of the enzyme, indicating successful docking.

Figure 5.2: Binding orientation of the docked ligand inside the ACE active site.

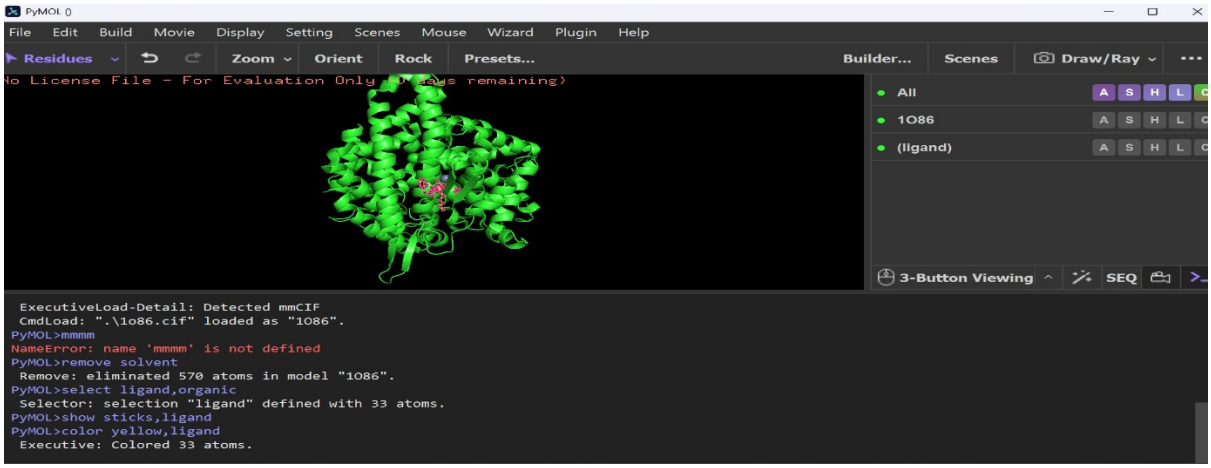
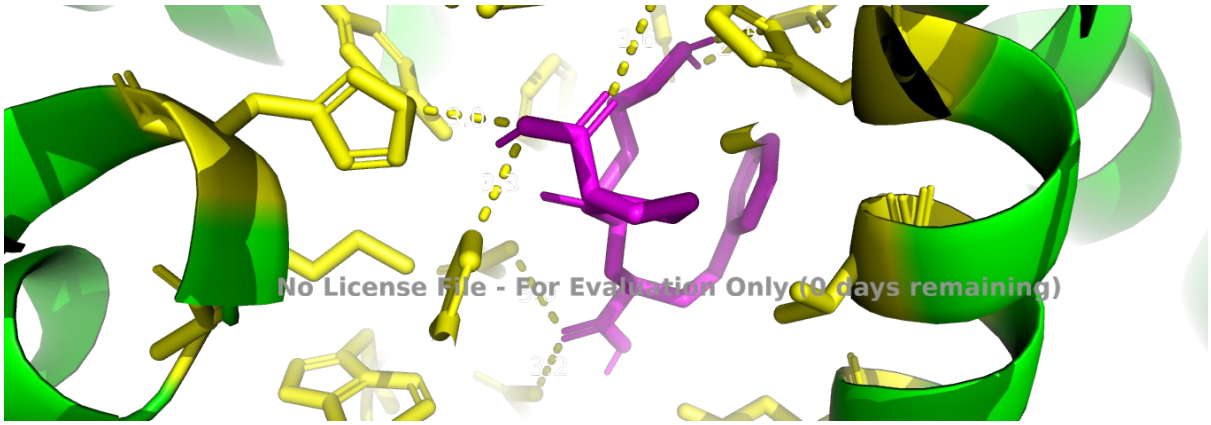


Figure 5.3: Binding interactions of Lisinopril with ACE active site residues.



Lisinopril exhibited multiple hydrogen bond interactions within the catalytic pocket, confirming its strong affinity towards ACE.

Figure 5.4: Molecular interactions of Captopril with ACE.

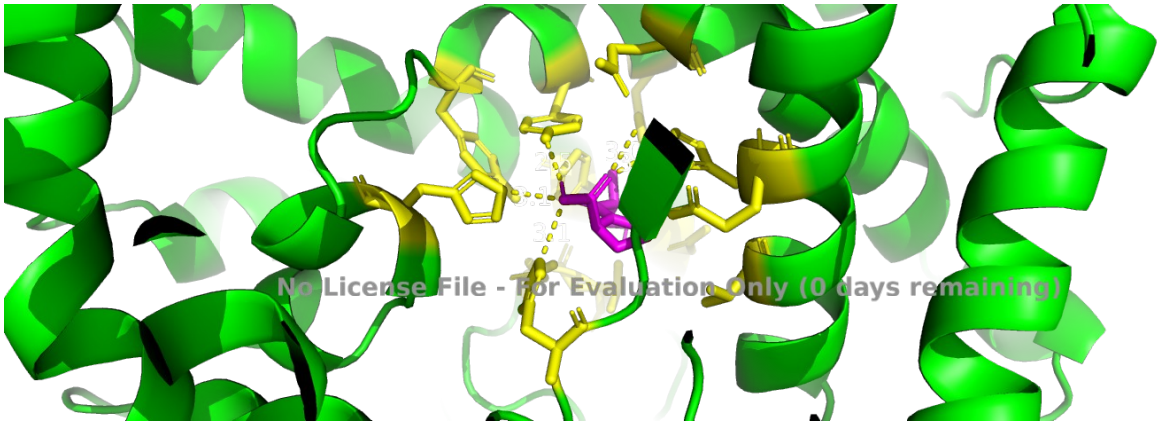


Figure 5.5 : Enalapril–ACE interaction

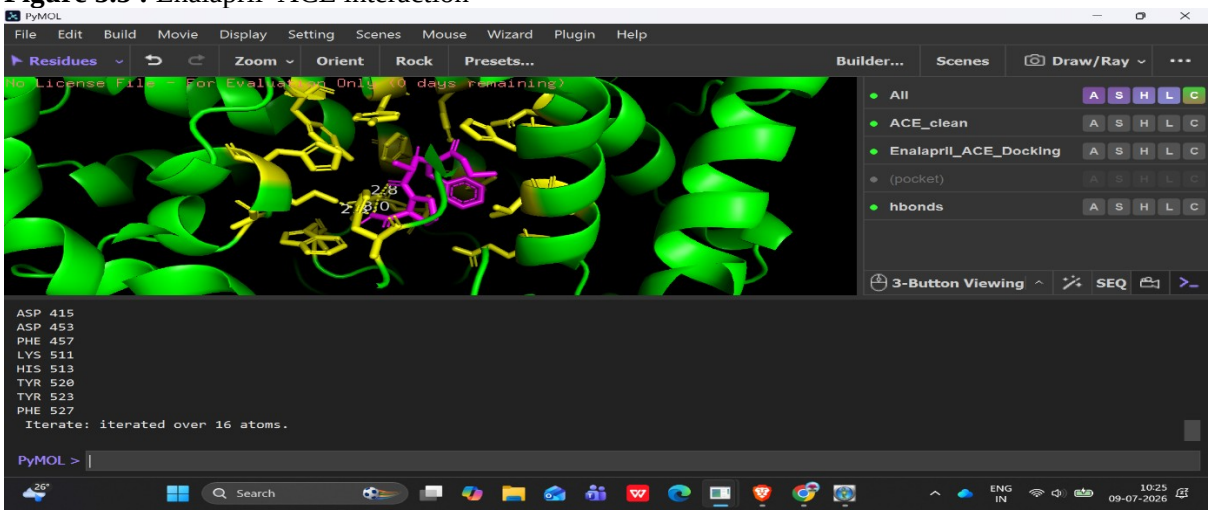
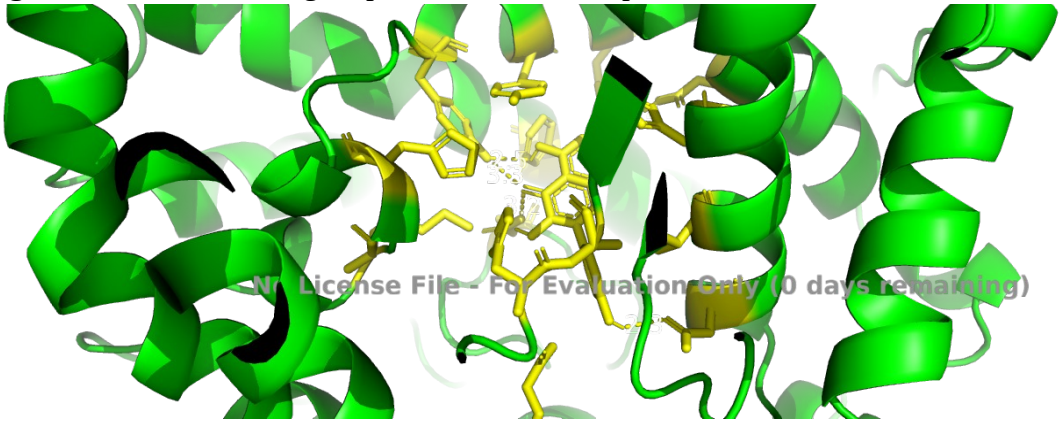


Figure 5.6: Binding pose of Kaempferol in the ACE active site



Kaempferol occupied the same catalytic region as the reference inhibitor, indicating its potential inhibitory activity.

Figure 5.7: Molecular interaction of Isoquercetin with ACE.

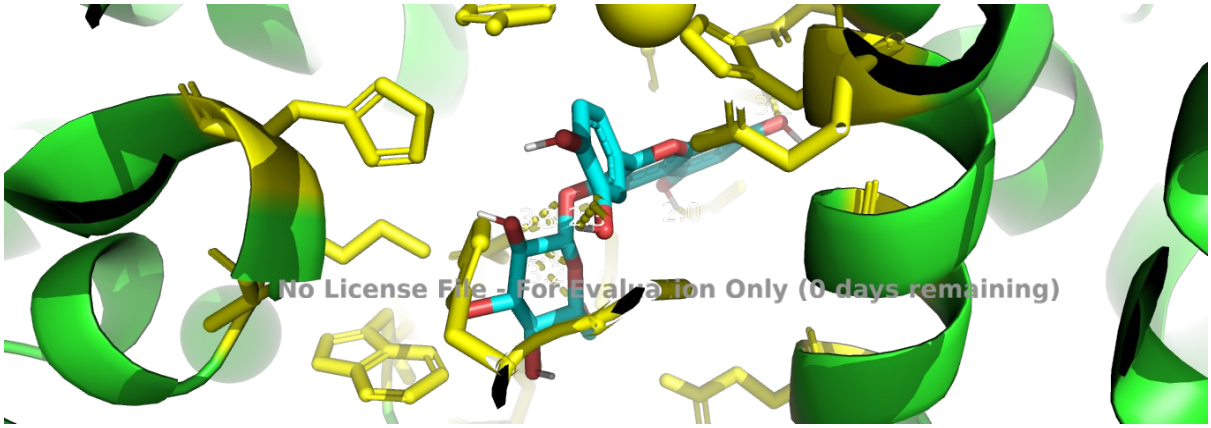


Figure 5.8: Binding conformation of Rutin inside the ACE binding pocket.

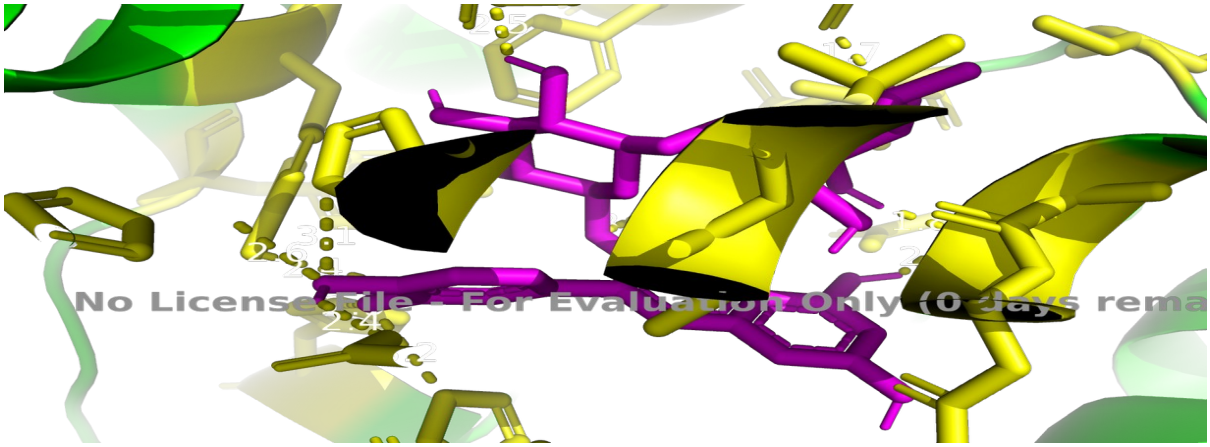


Figure 5.9: Docked complex of Astragalin with ACE.

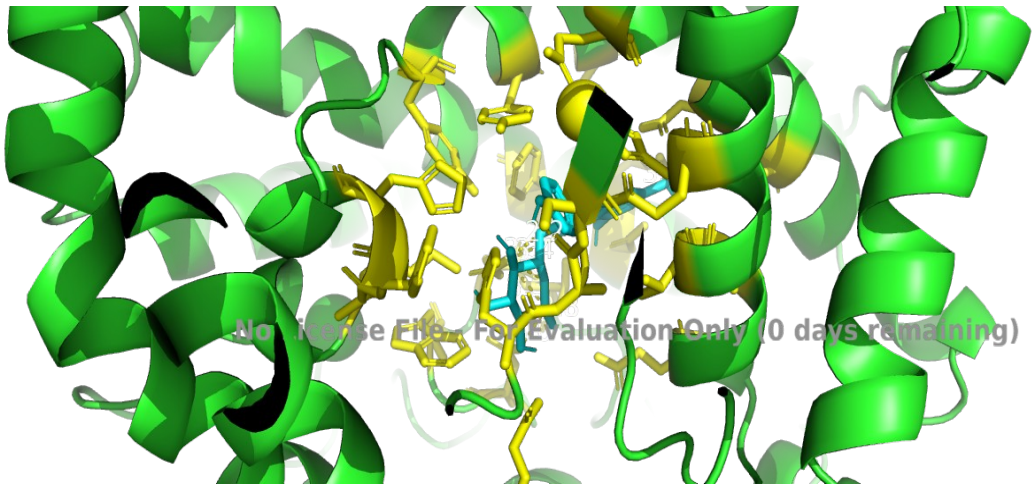


Figure 5.10: Docking interaction of Chlorogenic acid with ACE.

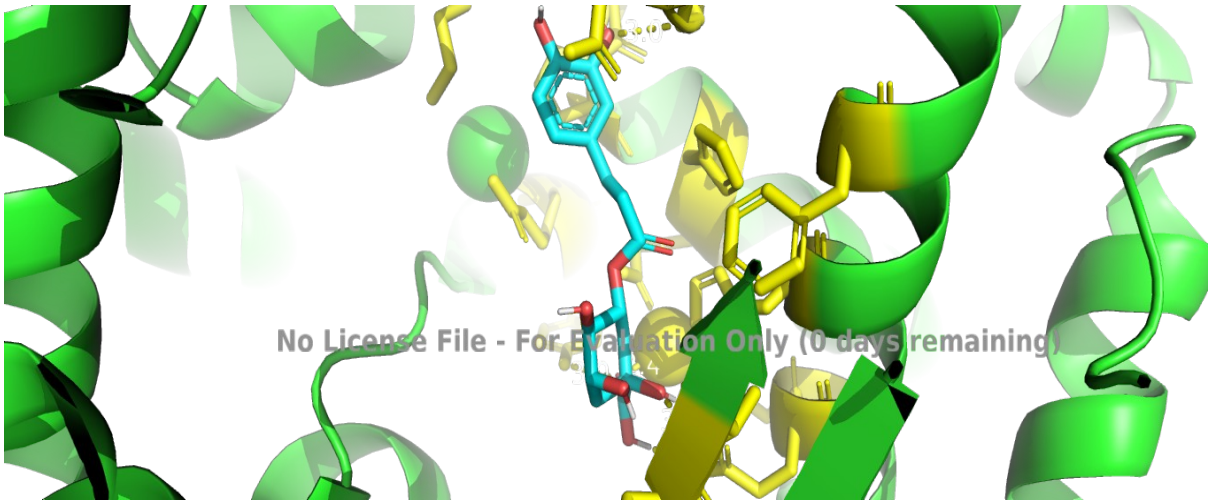


Figure 5.11: Docking interaction of Niaziminic with ACE.

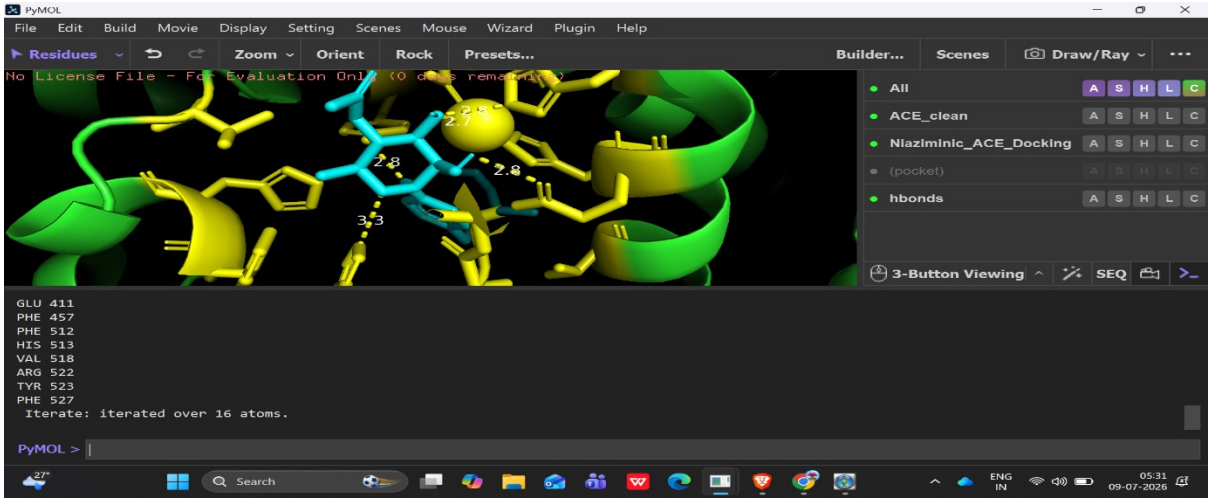
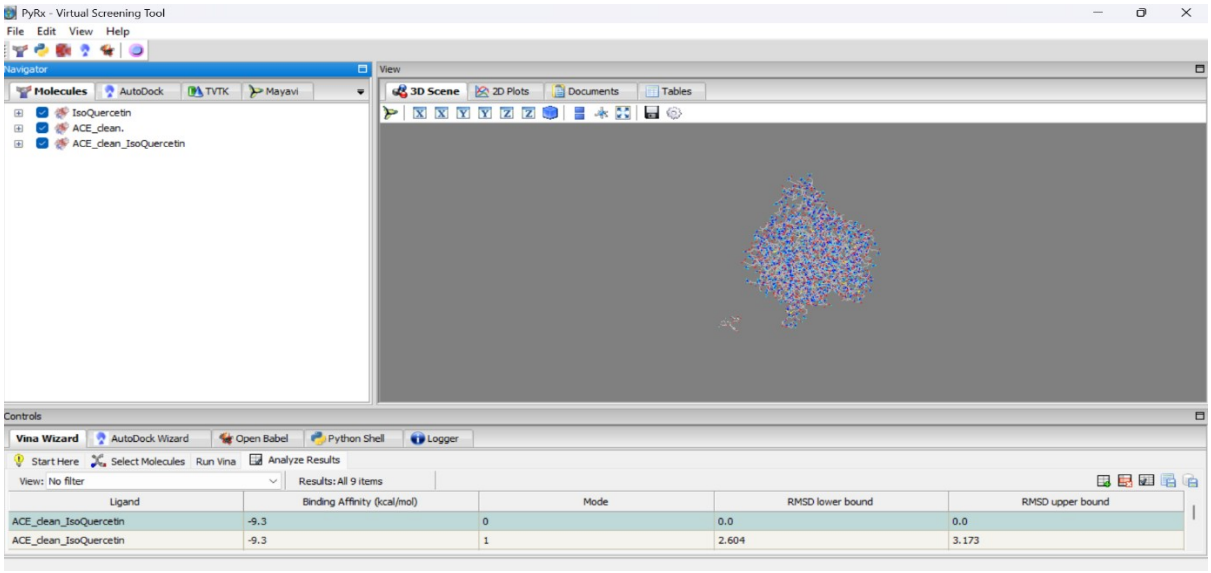


Figure 5.12: Docking results obtained using PyRx (AutoDock Vina), showing the binding affinity of Isoquercetin with ACE.



Molecular docking performed using AutoDock Vina predicted a binding affinity of **−9.3 kcal/mol** for Isoquercetin, indicating a strong interaction with the ACE binding pocket.

Table 5.1: Binding affinity of Moringa oleifera phytochemicals and reference ACE inhibitors against Angiotensin-Converting Enzyme (ACE).

S. No.	Ligand	Category	Binding Affinity (kcal/mol)	Remarks
1	Isoquercetin	Moringa phytochemical	**−9.3**	Highest binding affinity among tested phytochemicals
2	Rutin	Moringa phytochemical	**−9.2**	Strong interaction with ACE active site
3	Astragalin	Moringa phytochemical	**−8.8**	Stable binding with multiple interactions
4	Kaempferol	Moringa phytochemical	**−8.4**	Good binding affinity
5	Chlorogenic Acid	Moringa phytochemical	**−7.9**	Moderate binding affinity
6	Niaziminic Acid	Moringa phytochemical	**−7.6**	Moderate interaction with catalytic pocket
7	Lisinopril	FDA-approved ACE inhibitor	**−8.8**	Reference inhibitor
8	Enalapril	FDA-approved ACE inhibitor	**−8.1**	Reference inhibitor
9	Captopril	FDA-approved ACE inhibitor	**−6.0**	Reference inhibitor

Table 5.1: Presents the molecular docking scores obtained using AutoDock Vina through the PyRx platform. Isoquercetin exhibited the highest binding affinity (−9.3 kcal/mol), followed by Rutin (−9.2 kcal/mol), suggesting strong interactions with the active site of ACE. Several

phytochemicals demonstrated binding affinities comparable to or better than the FDA-approved ACE inhibitors, indicating their potential as natural ACE inhibitors.

6. Results and Discussion

6.1 Molecular Docking Analysis

Molecular docking was performed using AutoDock Vina implemented through the PyRx Virtual Screening Tool to investigate the binding affinity of selected *Moringa oleifera* phytochemicals towards Angiotensin-Converting Enzyme (ACE). FDA-approved ACE inhibitors, namely Lisinopril, Enalapril, and Captopril, were included as reference compounds for comparison. The binding affinity values obtained from molecular docking are summarized in Table 5.1.

Among all the screened phytochemicals, **Isoquercetin** exhibited the highest binding affinity (**−9.3 kcal/mol**), indicating a strong interaction with the active site of ACE. Other phytochemicals such as Rutin, Astragalin, Kaempferol, Chlorogenic Acid, and Niaziminic Acid also demonstrated favorable binding energies. Several phytochemicals displayed binding affinities comparable to or better than those of the reference inhibitors, suggesting their potential as natural ACE inhibitors.

The PyMOL interaction analysis revealed that the docked ligands occupied the catalytic binding pocket of ACE and formed multiple hydrogen bonds and hydrophobic interactions with important amino acid residues, including Asp453, Phe457, His513, Tyr520, Tyr523, Ser526, Phe527, and Gln530. These interactions contribute significantly to the stability of the protein–ligand complex.

Overall, the docking study indicates that selected phytochemicals from *Moringa oleifera* possess promising inhibitory potential against ACE and warrant further computational and experimental investigation.

6.2 Molecular Dynamics Simulation Analysis

To evaluate the stability of the docked protein–ligand complex under physiological conditions, Molecular Dynamics (MD) simulation was performed using GROMACS 2025.4. The simulation was carried out for **200 ps** after energy minimization and system equilibration. Various trajectory analyses, including RMSD, RMSF, Radius of Gyration (Rg), Hydrogen Bond analysis, and Solvent Accessible Surface Area (SASA), were performed to assess the structural stability of the complex.

6.2.1 Root Mean Square Deviation (RMSD)

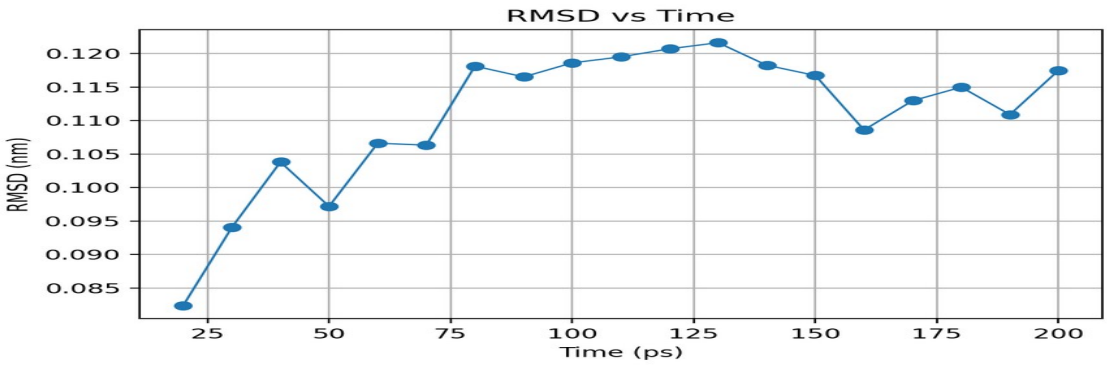


Figure 6.1: **RMSD profile of the ACE–ligand complex during the 200 ps Molecular Dynamics simulation**

The RMSD plot measures the overall structural deviation of the protein backbone during the simulation with respect to the initial structure. Throughout the 200 ps simulation, the RMSD values ranged from **0.082 nm to 0.121 nm**, with the final RMSD recorded at **0.117 nm**.

The RMSD initially increased during the first few frames as the system adjusted to the simulation environment. Subsequently, the fluctuations remained minimal, indicating that the protein–ligand complex rapidly achieved equilibrium and maintained structural stability throughout the simulation. The consistently low RMSD values suggest that ligand binding did not induce significant conformational changes in the ACE protein.

6.2.2 Root Mean Square Fluctuation (RMSF)

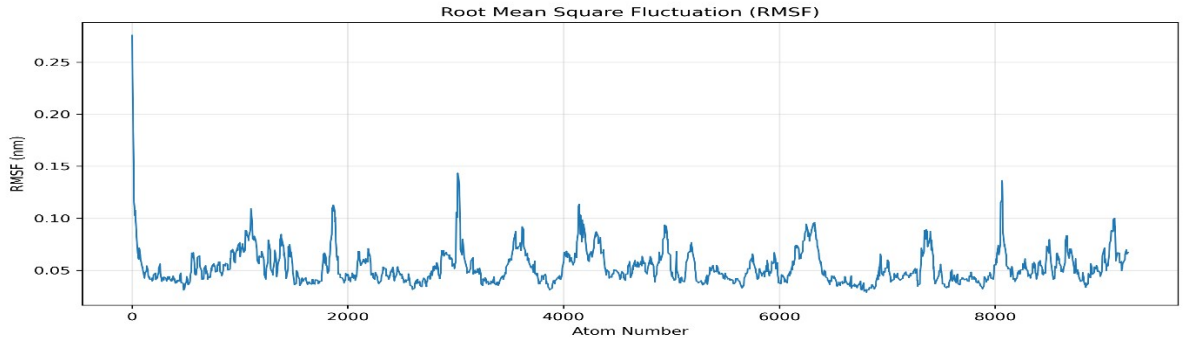


Figure 6.2: RMSF profile of ACE residues during Molecular Dynamics simulation.

The RMSF analysis evaluates the flexibility of individual amino acid residues throughout the simulation. Most residues exhibited low fluctuation values, generally remaining below **0.10 nm**, indicating limited atomic mobility. Slightly higher fluctuations were observed in loop regions and terminal residues, which is expected due to their inherent flexibility.

The catalytic residues involved in ligand binding remained relatively stable, confirming that the ligand maintained consistent interactions within the active site throughout the simulation.

6.2.3 Radius of Gyration (Rg)

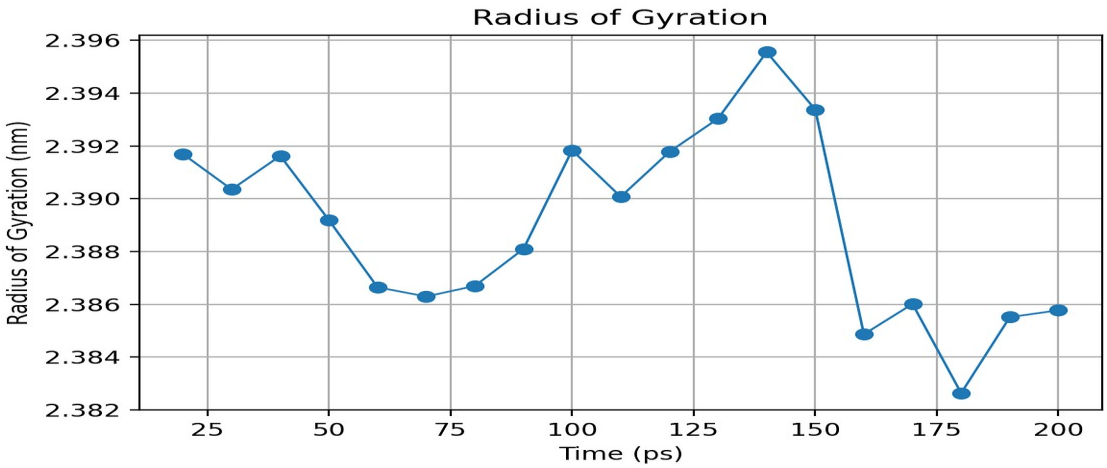


Figure 6.3: Radius of Gyration (Rg) of the ACE–ligand complex during the simulation

The Radius of Gyration (Rg) reflects the compactness of the protein structure during simulation. The Rg values remained nearly constant throughout the simulation, fluctuating only between approximately **2.383 nm and 2.396 nm**.

The absence of significant variation in Rg indicates that the overall tertiary structure of ACE remained compact and structurally stable. This observation further supports the structural integrity of the protein–ligand complex.

6.2.4 Hydrogen Bond Analysis

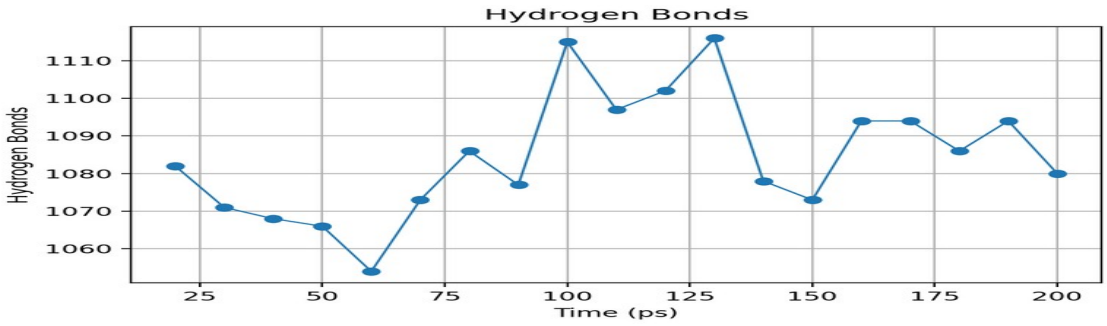


Figure 6.4: Variation in the number of hydrogen bonds formed during Molecular Dynamics simulation.

Hydrogen bonds play a crucial role in stabilizing protein–ligand interactions. During the simulation, the number of hydrogen bonds remained relatively constant, fluctuating between approximately **1054 and 1116**.

The stable hydrogen bond profile indicates persistent intermolecular interactions throughout the simulation, thereby contributing to the overall stability of the ACE–ligand complex.

6.2.5 Solvent Accessible Surface Area (SASA)

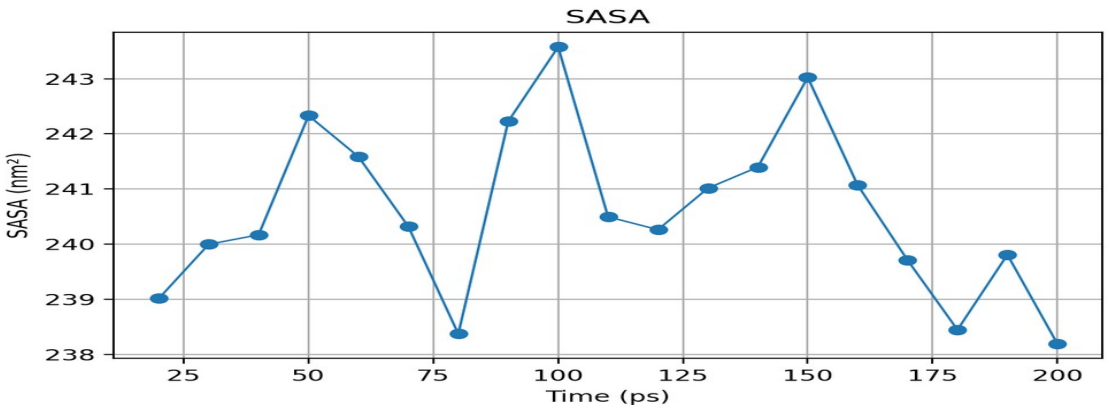


Figure 6.5: Solvent Accessible Surface Area (SASA) profile of the ACE–ligand complex.

The SASA analysis estimates the solvent-exposed surface area of the protein. During the simulation, SASA values remained relatively stable, ranging between approximately **238 and 244 nm²**.

The limited variation in SASA suggests that no major unfolding or structural disruption occurred during the simulation. This finding further confirms the structural stability of the ACE–ligand complex under simulated physiological conditions.

Overall Discussion

The combined molecular docking and molecular dynamics simulation results indicate that the selected *Moringa oleifera* phytochemicals exhibit promising interactions with Angiotensin-Converting Enzyme (ACE). Docking analysis identified compounds with strong binding affinities and favorable interactions within the catalytic pocket, while the MD simulation confirmed that the protein–ligand complex remained structurally stable over the 200 ps simulation. Low RMSD and RMSF values, a nearly constant Radius of Gyration, stable hydrogen bond formation, and minimal SASA fluctuations collectively demonstrate that the complex maintained its structural integrity throughout the simulation. These findings support the potential of *Moringa oleifera* phytochemicals as candidates for further investigation as natural ACE inhibitors.

7. Skills Gained

The eight-week internship on **Computer-Aided Drug Design (CADD)** provided extensive hands-on experience in computational drug discovery, molecular modeling, molecular docking, and molecular dynamics simulation. The internship significantly enhanced both technical and analytical skills required in modern bioinformatics and pharmaceutical research.

7.1 Technical Skills

Molecular Docking (AutoDock Vina)

- Performed molecular docking using AutoDock Vina through the PyRx platform.

- Compared the binding affinities of *Moringa oleifera* phytochemicals with FDA-approved ACE inhibitors.
- Analyzed binding energies and identified the best-performing ligands.

Protein Structure Preparation

- Retrieved the crystal structure of Human Angiotensin-Converting Enzyme (ACE) from the Protein Data Bank.
- Prepared the protein using PyMOL by removing water molecules, heteroatoms, and native ligands.
- Generated a docking-ready protein structure.

Ligand Preparation

- Downloaded phytochemical structures from PubChem.
- Converted ligand structures from SDF to PDB format using Open Babel.
- Optimized ligand structures for molecular docking.

Docking Protocol Validation (Redocking/RMSD)

- Validated the docking protocol through redocking of the native ligand.
- Evaluated docking accuracy using RMSD analysis.

Protein–Ligand Interaction Analysis

- Visualized docking poses using PyMOL.
- Analyzed hydrogen bonds, hydrophobic interactions, and active-site residues using Discovery Studio Visualizer.

ADMET Prediction

- Performed drug-likeness analysis using SwissADME.
- Evaluated pharmacokinetic and toxicity properties using pkCSM.
- Interpreted Lipinski's Rule of Five and other ADMET parameters.

Molecular Dynamics Simulation (GROMACS)

- Performed Molecular Dynamics simulation using GROMACS 2025.4 in Ubuntu (WSL).
- Conducted energy minimization, NVT equilibration, NPT equilibration, and production MD simulation.
- Analyzed RMSD, RMSF, Radius of Gyration, Hydrogen Bonds, and SASA to evaluate complex stability.

Linux Command Line

- Learned basic Linux commands for file handling and execution of GROMACS workflows.
- Managed simulation files and directories in the Ubuntu terminal.

Scientific Data Analysis

- Interpreted docking scores and molecular dynamics trajectories.
- Organized computational data into tables, figures, and graphs.
- Prepared scientific documentation and research-style reports.

Software and Tools

- PyMOL
- PyRx
- AutoDock Vina
- Discovery Studio Visualizer
- Open Babel
- GROMACS 2025.4
- SwissADME
- pkCSM
- Ubuntu (WSL)
- Microsoft Excel
- GitHub (basic usage)

8. Challenges Faced and Solutions

- Missing .gro files during GROMACS setup.
- Errors during NVT and NPT equilibration.
- Incorrect .mdp parameters requiring troubleshooting.
- Laptop shutdown during the production MD run; simulation successfully resumed using checkpoint (.cpt) files.
- Linux command-line troubleshooting.
- Understanding trajectory analysis (RMSD, RMSF, Rg, SASA, Hydrogen Bonds).

9. Conclusion

The present study successfully employed Computer-Aided Drug Design (CADD) approaches to investigate the interaction of selected *Moringa oleifera* phytochemicals with Human Angiotensin-Converting Enzyme (ACE). Molecular docking identified several phytochemicals with favorable binding affinities comparable to FDA-approved ACE inhibitors, suggesting their potential as natural antihypertensive candidates.

The docking results were further supported through Molecular Dynamics simulation using GROMACS, where the protein–ligand complex demonstrated excellent structural stability throughout the 200 ps simulation. RMSD, RMSF, Radius of Gyration, Hydrogen Bond analysis, and SASA collectively confirmed the stability and integrity of the complex under simulated physiological conditions.

In addition to the scientific outcomes, this internship provided valuable practical experience in protein preparation, ligand preparation, molecular docking, docking validation, ADMET prediction, molecular dynamics simulation, trajectory analysis, Linux-based computational workflows, and scientific data interpretation. The project enhanced understanding of computational drug discovery and strengthened technical skills essential for future research in bioinformatics, structural biology, and pharmaceutical sciences.

Overall, the findings indicate that selected *Moringa oleifera* phytochemicals exhibit promising interactions with ACE and may serve as potential lead compounds for further in vitro, in vivo, and clinical investigations.

10. References

Use **APA style** or your institute's preferred format. Include references such as:

1. Trott, O., & Olson, A. J. (2010). AutoDock Vina: Improving the speed and accuracy of docking. *Journal of Computational Chemistry*, 31(2), 455–461.
2. Abraham, M. J., et al. (2015). GROMACS: High-performance molecular simulations through multi-level parallelism. *SoftwareX*, 1–2, 19–25.
3. Berman, H. M., et al. (2000). The Protein Data Bank. *Nucleic Acids Research*, 28(1), 235–242.
4. Kim, S., et al. (2023). PubChem database. *Nucleic Acids Research*.
5. Daina, A., Michielin, O., & Zoete, V. (2017). SwissADME. *Scientific Reports*, 7, 42717.
6. Pires, D. E. V., et al. (2015). pkCSM. *Journal of Medicinal Chemistry*, 58(9), 4066–4072.
7. Schrödinger, LLC. PyMOL Molecular Graphics System.
8. BIOVIA Discovery Studio Visualizer User Guide.
9. Open Babel Documentation.
10. PyRx Virtual Screening Tool Documentation.

11. Appendix

Appendix A: Linux Commands Used

- gmx pdb2gmx
- gmx editconf
- gmx solvate
- gmx genion
- gmx grompp
- gmx mdrun
- gmx trjconv
- gmx rms
- gmx rmsf
- gmx gyrate
- gmx hbond
- gmx sasa

Appendix B: Workflow Screenshots

- Protein preparation in PyMOL
- Ligand preparation
- PyRx docking setup
- Discovery Studio interaction analysis
- GROMACS simulation workflow

Appendix C: Molecular Dynamics Results

- RMSD graph
- RMSF graph
- Radius of Gyration graph
- Hydrogen Bond graph
- SASA graph

Appendix D: Additional Figures and Tables

- Docking interaction images
- Binding affinity table
- ADMET summary table